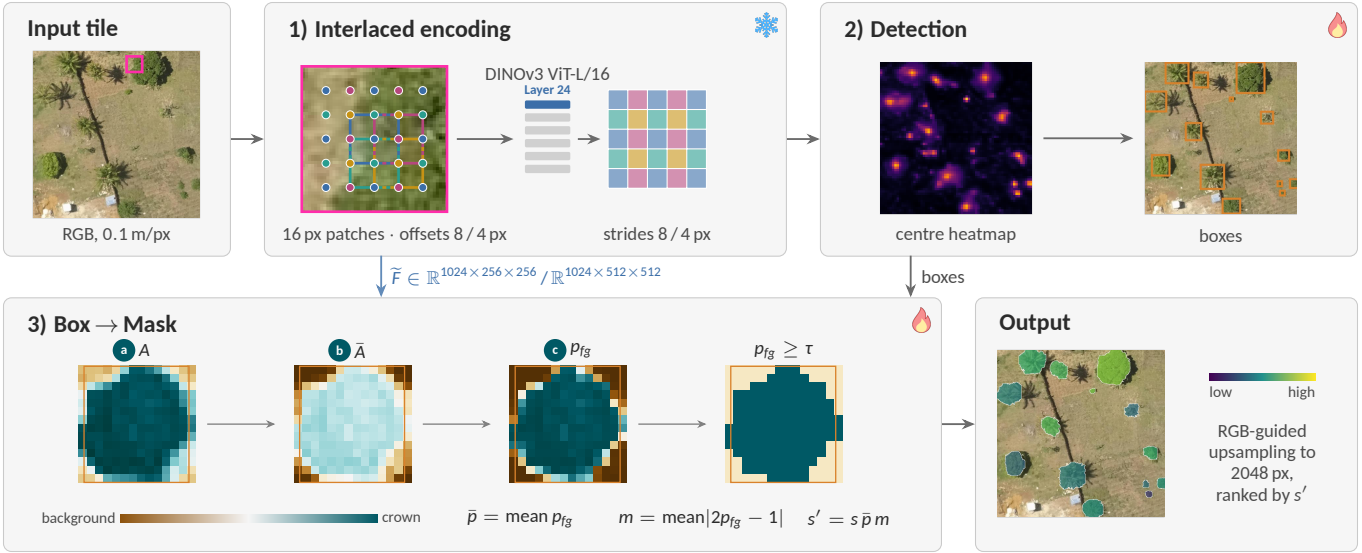

Input tile
RGB, 0.1 m/px
1) Interlaced encoding
DINOv3 ViT-L/16
Layer 24
16 px patches · offsets 8 / 4 px
strides 8 / 4 px
2) Detection
centre heatmap
boxes
$\tilde{F} \in \mathbb{R}^{1024\times256\times256}$ / $\mathbb{R}^{1024\times512\times512}$
boxes
3) Box → Mask
a A
b $\bar{A}$
c p_{fg}
$p_{fg} \geq \tau$
background
crown
$\bar{p} = \text{mean}\, p_{fg}$
$m = \text{mean}|2p_{fg} - 1|$
$s' = s\,\bar{p}\,m$
Output
low
high
RGB-guided upsampling to 2048 px, ranked by s'